Exp 8:

Mainactivity.java:

```
package com.example.ch8;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.io.BufferedReader;
import java.io.File;
import java.io.FileReader;
import java.io.FileWriter;

public class MainActivity extends AppCompatActivity {

    EditText roll, name, dept, searchRoll, result;
    Button addBtn, findBtn, showBtn;
    File file;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        roll = findViewById(R.id.roll);
        name = findViewById(R.id.name);
        dept = findViewById(R.id.dept);
        searchRoll = findViewById(R.id.searchRoll);
        result = findViewById(R.id.result);

        addBtn = findViewById(R.id.addBtn);
        findBtn = findViewById(R.id.findBtn);
        showBtn = findViewById(R.id.showBtn);

        File path = getExternalFilesDir(null);
        file = new File(path, "student.txt");

        // ADD STUDENT
        addBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                String r = roll.getText().toString();
                String n = name.getText().toString();
                String d = dept.getText().toString();

                if (r.isEmpty() || n.isEmpty() || d.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Enter all details",
Toast.LENGTH_SHORT).show();
                    return;
```

```
                }

                try {

                    FileWriter writer = new FileWriter(file, true);
                    writer.write(r + "," + n + "," + d + "\n");
                    writer.close();

                    Toast.makeText(MainActivity.this, "Student Saved",
Toast.LENGTH_SHORT).show();

                    roll.setText("");
                    name.setText("");
                    dept.setText("");

                } catch (Exception e) {
                    Toast.makeText(MainActivity.this, "Write Error: " +
e.getMessage(), Toast.LENGTH_LONG).show();
                }
            }
        });

        // FIND STUDENT
        findBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                String search = searchRoll.getText().toString();
                String line;
                boolean found = false;

                try {

                    BufferedReader br = new BufferedReader(new
FileReader(file));

                    while ((line = br.readLine()) != null) {

                        String[] data = line.split(",");

                        if (data[0].equals(search)) {

                            result.setText(data[0] + "," + data[1] + "," +
data[2]);

                            found = true;
                            break;
                        }
                    }

                    br.close();

                    if (!found) {
                        result.setText("Student Not Found");
                    }

                } catch (Exception e) {
```

```
                    Toast.makeText(MainActivity.this, "Read Error: " +
e.getMessage(), Toast.LENGTH_LONG).show();
                }
            }
        });

        // SHOW ALL STUDENTS
        showBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                StringBuilder data = new StringBuilder();
                String line;

                try {

                    BufferedReader br = new BufferedReader(new
FileReader(file));

                    while ((line = br.readLine()) != null) {

                        String[] s = line.split(",");

                        data.append(s[0]).append(",")
                                .append(s[1]).append(",")
                                .append(s[2]).append("\n");
                    }

                    br.close();

                    result.setText(data.toString());

                } catch (Exception e) {
                    Toast.makeText(MainActivity.this, "Error: " +
e.getMessage(), Toast.LENGTH_LONG).show();
                }
            }
        });
    }
}
```

AndroidManifest.XML:

```
<?xml version="1.0" encoding="utf-8"?>

<manifest
xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission
android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
    <uses-permission
android:name="android.permission.READ_EXTERNAL_STORAGE"/>

    <application
        android:allowBackup="true"
        android:label="@string/app_name"
        android:theme="@style/Theme.Ch8">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action
android:name="android.intent.action.MAIN"/>
                <category
android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>

        </activity>

    </application>

</manifest>
```

Output:

Student Details

Roll Number

Student Name

Department

Add Student

101

Find Student

Show All Students

101,Varshini,CSE
102,Vaithee,ECE
103,Sri,EEE